

Achieva

Executive Summary

Clients

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Student Consulting Team

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Background

Achieva is a non-profit organization based in Southwestern Pennsylvania dedicated to advocating for and providing comprehensive support services to people with disabilities and their families. Achieva's ultimate goal is to foster inclusion and independence for its community members through vocational support, disability advocacy, and specialized home and community-based services.

Project Description

Project Opportunity

Achieva identified a critical service gap for its members in Westmoreland County, where a lack of reliable public transportation created a significant barrier to employment for people with disabilities. Previously, many clients relied on a single, inconsistent bus route that often failed to accommodate workplace schedules, forcing them to rely heavily on family members for daily commutes or spend limited income on third-party rideshare services that could not always meet their specific accessibility needs. Recognizing this need, Achieva sought out our student team to launch a rideshare program to provide free, dependable transportation to and from work for its members. The opportunity involved developing a custom web application to manage this new initiative- starting with a pilot in Westmoreland County with a long-term vision of expanding the service throughout the greater Pittsburgh area. This system aims to provide a life-changing tool for passengers to achieve greater professional and personal independence.

Project Vision

To address this need, we created a rideshare platform that is easy to use for passengers of varying abilities, provides optimized navigation for drivers, and reduces the manual administrative burden for Achieva staff. We hoped that by integrating automated routing logic, we would save the transportation coordinators significant time, allowing them to focus on expanding service capacity. Ultimately, the vision was to empower passengers to hold jobs independently by providing a reliable, low-cost alternative to commercial transportation services.

Project Outcomes

The team developed a custom rideshare application for Achieva using a relational SQL database and a modern web framework. The application allows passengers to autonomously schedule work-related transit- directly supporting their professional independence and reducing reliance on family members or expensive third-party services. Additionally, the system is integrated with Google Fleet Routing API that generates multi-stop navigation links, eliminating the need for manual coordinate entry and allowing drivers to receive drive information and update trip statuses all in one page. Lastly, administrators are provided with a centralized management dashboard- allowing them to edit passengers/drivers and monitor daily trips as they happen. By implementing this rideshare application, Achieva now has a scalable solution to provide free and reliable transportation to their community members.

Project Deliverables

The team delivered a fully functional rideshare application, aptly named ***AchievaPath***. The application incorporates all essential accessibility requirements and will serve as the official platform for Achieva's workplace transportation program. The system includes a route optimization engine that seamlessly generates navigation links for all daily trips. Furthermore, we provided Achieva with comprehensive documentation and user guides for Administrators, Drivers, and Passengers to ensure long-term system sustainability.

Recommendations

To help Achieva sustain its progress, the team has migrated the application to a permanent Virtual Machine to ensure 24/7 availability for drivers and passengers. The team recommends that Achieva perform a final administrative data cleanse of all testing records to ensure a clean environment to add real passenger data. Additionally, Achieva should utilize the newly implemented data-tracking features to generate monthly mileage reports- facilitating easier financial reporting and grant compliance. It is also recommended that the passenger listing be updated and audited on a weekly basis to ensure all accommodation requirements and addresses remain current. Finally, the team suggests designating a technical lead (Ryan Jackson) to oversee the system's security protocols and manage API key access to ensure the long-term integrity of the platform

Student Consulting Team

Ashley Liu served as the Lead Backend Developer. She is a third-year student double majoring in Information Systems and Computer Science. This summer, she is interning as a software development engineer at Amazon.

Kayla Edwards served as the Project Manager and a Backend Developer. She is a third-year student majoring in Information Systems with a Minor in Human Computer Interaction. This summer, she is interning at Bank of America as a Global Technology Business Analyst.

Victoria Solsky served as the Frontend Lead. She is a fourth-year student majoring in Information Systems with a Minor in Human Computer Interaction. Post graduation, she will be moving back home and spending time with family and friends, as well as traveling and will start work in the fall.